CSE 4088 Introduction to Machine Learning
Fall 2020-2021
MIDTERM EXAM

Start 15:00 – End 17:00

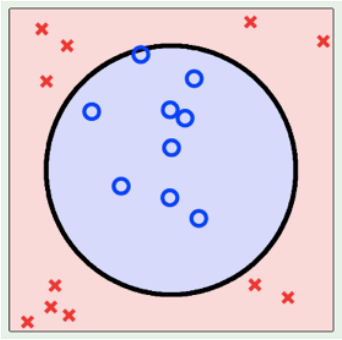
Name:
Student ID:

Q1	**/ 35**
Q2	**/ 30**
Q3	**/ 35**
Total	**/ 100**

- There are 3 questions and 4 pages in this handout.
- Please use A4 size white blank sheets to write your answers with handwriting. You can print this booklet and write your answers on it if you have a printer.
- Please write your name and student ID at the top of all your answer sheets. Please copy the above grading table to the first page as well.
- Please copy the sentences below to the first page of your answer sheet in handwriting and sign.
  HONOR CODE: "*On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for this homework. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues*."< your signature>
- You have 2 hours to complete your answers and upload them to UES in a single pdf document. The document should be named as "**name_surname.pdf**". You should pay attention to the following:
  - Use a scanner or a **scanner application on your mobile phone** (Adobe Scan etc.) to scan your answers into a single pdf file. **Do not take photos** of the pages.
  - Scan each page one by one in upright **portrait orientation** (NOT landscape).
  - Make sure that all pages are clear and **readable**.
  - Do not delay your submission to the last minute as there may be unexpected problems. Try to begin uploading your file at least 10 minutes before the deadline.
  - You can also upload your answers to Google Classroom course web page if and only if a problem occurs with UES. **Do NOT** upload to both UES and Google Classroom.
- **Chrome browser** is recommended to minimize any problems.
- SHOW ALL YOUR WORK CLEARLY AND STEP BY STEP TO RECEIVE CREDIT. Good luck!

Question 1:

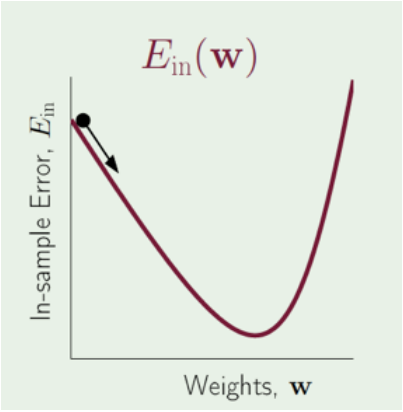
Suppose we have the two-class and two-dimensional data set given below, which is not linearly separable. The origin is at the center of the figure. Propose **two different solutions** to classify this data set. In each case, give the details of your method as much as you can using mathematical equations as necessary.



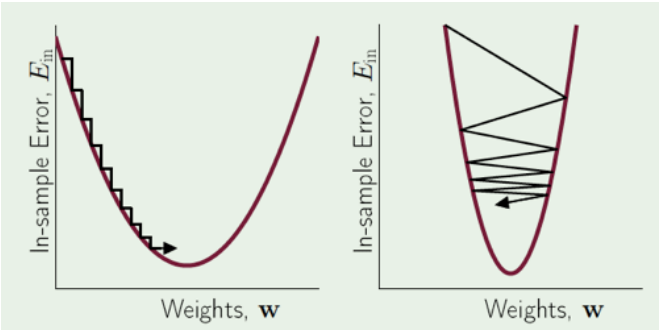
Question 2 – Gradient Descent

Suppose we are given a training dataset consisting of N samples: $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)$, where the input feature vector is $\mathbf{x}$ and the output y is the correct output. Suppose we want to learn a hypothesis $h(\mathbf{x})$, so that we aim to achieve the correct decision $h(\mathbf{x}_n) = y_n$ on all the training examples. Let $h(\mathbf{x})$ be parametrized by a vector $\mathbf{w}$ and let $e(h(\mathbf{x}_n), y_n)$ denote the error for one of the n samples.

(a) Write an algorithm that uses the **batch** gradient descent algorithm to estimate the parameter vector **w.** First, express $E_{in}(\mathbf{w})$ in terms of $e(h(\mathbf{x}_n), y_n)$. Explain how we can use batch gradient descent to minimize $E_{in}(\mathbf{w})$. You may use the below diagram in your explanation. Explain all the terms and constants that you introduce and comment on the range of values that they can take. Include the initialization and the stopping criterion, as well.



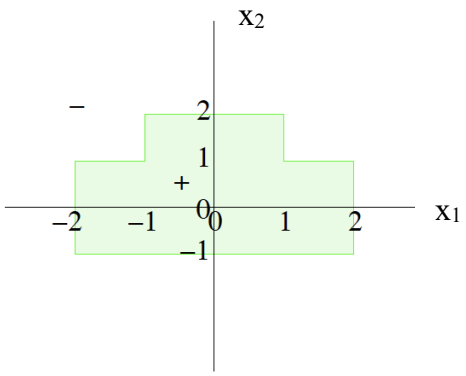
(b) Explain the reasons for the two problematic convergence behaviors illustrated in the figures below for batch gradient descent. How can we solve the problem in each case?



(c) Write an algorithm that uses the **stochastic** gradient descent algorithm to estimate the parameter vector **w.** What are the advantages of stochastic gradient descent over batch gradient descent?

Question 3 – Neural Networks

Implement the decision function below for a binary classification problem using a 3-layer perceptron. The shaded region denotes where the output should be +1. Draw the 3 layer perceptron and show all the weights.



Note: Use the following perceptron model:

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^{\mathbf{T}}\mathbf{x})$$

where $h(\mathbf{x})$ is the output hypothesis (which can be +1 or -1), $\mathbf{w} = [w_0 \quad w_1 \quad w_2]^T$ denotes the weights, and $\mathbf{x} = [1 \quad x_1 \quad x_2]^T$ denotes the input vector.

Hint: Describe the target region as a combination of perceptrons which implement linear decision boundaries. Combine the perceptrons using Boolean operations. Boolean AND and OR operations (for two inputs) can be implemented by the perceptron as follows, where z_1 and z_2 can take the values +1 or -1.

$$\text{OR}(z_1, z_2) = \text{sign}(z_1 + z_2 + 1.5)$$
$$\text{AND}(z_1, z_2) = \text{sign}(z_1 + z_2 - 1.5)$$